Prediction 1: Single-Bond Defect Topological Response Experiment Report

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Abstract

This experiment report documents the complete process of testing Prediction 1 of the Electron Delocalization Relation Network (EDRN) framework through single-bond defect scanning on the one-dimensional half-filled Hubbard model. Under the methodological guidance of the Crystal Vein Philosophy—practice intervention theory, rigorous prevention of the tool-rationality paradox, firm opposition to "alignment", and "only diagnose, not prescribe"—we fixed the chain length L and varied the hopping strength of a single bond at the chain center (from $0.5t$ to $1.5t$). For $L = 40$ and $L = 60$, we measured the spin gap Δ_s, the prefactor $A = \Delta_s \times L$, the topological connectivity index C, and the charge gap ΔE. Ten independent data points consistently show that A undergoes a significant, monotonic, nonlinear variation with the local bond strength, with a change exceeding 500%. In contrast, C remains nearly constant (change <1%). The standard theoretical prediction that A is a bulk universal constant does not hold under the present experimental conditions. Charge gap measurements confirm that the system remains in the Mott insulating phase for all defect strengths, ruling out a metal-insulator transition. The A(defect) curves retain a consistent shape at two independent L values, passing the L-convergence test. This experiment provides the first direct numerical evidence for the "relation precedes entity" postulate within the EDRN framework in strongly correlated physics. All data, scripts, and the experimental protocol are publicly available on GitHub.

1 Scientific Question and Experimental Objective

1.1 Original Conjecture

In a Mott insulator, the prefactor A in the finite-size scaling form of the spin gap $\Delta_s = A(C)/L$ might be related to the network's topological connectivity index C. This conjecture originates from the core postulate of relation ontology—if reality is constituted by relation networks, then changing the network structure should alter the system's physical response. More precisely: **if "relation precedes entity" is a physically valid proposition, then the prefactor**

of spin excitations should not be a bulk universal constant, but should vary with changes in the network topology.

1.2 Lesson from the First Experiment (July 14, 2026)

Under single-variable variation (changing L) on a linear chain, C is a deterministic function of L:

$$C \approx 0.9745 - 0.624/L, \quad R^2 = 0.99999. \tag{1}$$

Consequently, the high R^2 correlation between A and C is a statistical inevitability—both A and C drift smoothly with L, and fitting one L-dependent quantity against another yields a high R^2 as a mathematical necessity, not a physical discovery (Drahos, 2026, GitHub Issue #1). After removing the $1/L$ trend from A, the explanatory power of C for the residuals is zero ($R^2 = 0.00$).

This lesson directly points to the correct testing scheme: **one must fix L and generate independent variation by changing the local topological structure, then test whether A follows that variation.**

1.3 Objective of This Experiment

Under fixed L, by varying the hopping strength of a single bond defect (from $0.5t$ to $1.5t$), we test the following two opposing predictions:

- **Standard theory prediction (entity priority):** A is a bulk universal constant, independent of local bond strength. The Heisenberg model and Luttinger liquid theory, under the assumption of a uniform chain, derive a constant spin-wave velocity v_s, and $A = \pi v_s$ should remain unchanged.
- **EDRN framework prediction (relation priority):** A varies with local bond strength. If reality is constituted by relation networks, then altering the local structure of the network—even the strength of a single bond—should change the system's global physical response. Relation precedes entity; structural change necessarily leads to response change.

1.4 Methodological Constraints

This experiment strictly adheres to the following principles, which are not decorative philosophical declarations but operational norms binding on every experimental design decision:

- **Rigorous prevention of the tool-rationality paradox:** All defect strengths use unified DMRG parameters. If a data point fails to converge, a sentinel value is recorded with the reason stated, without forced repair. No selective reporting or parameter fine-tuning is applied after data production to "align" the results with theory.
- **Firm opposition to "alignment":** This experiment does not presuppose the validity of either prediction. All data are presented side by side, in raw form. Regardless of whether the results show A unchanged (supporting standard theory), A varying (supporting the

EDRN framework), or unexpected outcomes, everything will be honestly recorded and reported.

- **Only diagnose, not prescribe:** This experiment solely diagnoses "whether the spin-gap prefactor A varies with local bond strength at fixed L". No advance judgment is made on "what this means", and no normative conclusions are drawn.
- **Practice intervention theory:** The introduction of a single-bond defect is an active intervention into the theoretical field. The chosen range of defect strengths and L values constitutes a specific cut into the real physical field. One must honestly examine the boundaries of this cut—clarifying which physical effects are captured by the current design and which may be systematically missed.

2 Experimental Design

2.1 Model and Parameters

Model: One-dimensional single-band Fermi-Hubbard chain, Hamiltonian:

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \tag{2}$$

Fixed parameters:

- $U/t = 4$ (strongly correlated Mott insulating regime)
- Half-filling ($\langle n \rangle = 1$)
- Open boundary conditions
- Bond dimension $\chi_{\max} = 100$
- SVD truncation tolerance svd_min $= 10^{-10}$
- Maximum number of sweeps max_sweeps $= 50$
- Minimum number of sweeps min_sweeps $= 2$

2.2 Method of Introducing the Single-Bond Defect

At the central bond of the chain (index $L//2 - 1$), the hopping strength is modified from the uniform value $1.0t$ to a specified value t_{defect}, while all other bonds remain at $1.0t$. Different degrees of local topological variation are generated by varying t_{defect}.

Defect strength sequence: $0.5t, 0.8t, 1.0t$ (uniform chain baseline), $1.2t, 1.5t$. This covers the full parameter range from a "weak link" to a "strong link". $0.5t$ approaches the extreme case of "almost severed", while $1.5t$ is an "enhanced link".

2.3 Scale Setting

Two independent scales: $L = 40$ and $L = 60$. The two L values are analyzed independently, without mutual alignment. If the variation of A maintains a consistent monotonic trend across both scales, it supports a bulk law. If the A values at $L = 60$ regress toward the uniform-chain value, it would indicate a finite-size transient.

2.4 Observables

1. **Spin gap $\Delta_s = E(S = 1) - E(S = 0)$:** Energy difference between the spin triplet and spin singlet ground states. The triplet initial state is constructed by flipping the spins at the chain center and its adjacent site (yielding $S_z = 2$), avoiding DMRG convergence back to the ground state.

2. **Prefactor $A = \Delta_s \times L$:** Dimensionless quantity after removing the leading-order $1/L$ size dependence of Δ_s. Standard theory predicts A approaches the constant πv_s in the thermodynamic limit.

3. **Topological connectivity index C:** Estimated from the off-diagonal elements of the single-particle density matrix, measuring the overall connectivity of the network.

4. **Charge gap $\Delta E = E(N) - E(N - 1)$:** Gate condition—confirms that the system remains in the Mott insulating phase (charge gap nonzero) for all defect strengths.

2.5 Gate Condition

All data points must pass the following gate conditions before entering the A(defect) analysis:

1. **Charge gap nonzero:** $\Delta E > 0$, confirming the system is in the Mott insulating phase.

2. **Spin gap satisfies $\Delta_s \sim 1/L$:** The product $\Delta_s \times 60$ at $L = 60$ should be close to $\Delta_s \times 40$ at $L = 40$, confirming the spin sector is gapless.

If the charge gap closes ($\Delta E \to 0$) at any defect strength, that data point is excluded—the system is no longer a Mott insulator and does not belong to the inspection scope of this experiment. This is not "bad data" but a "category error".

2.6 Computational Workflow

All calculations were performed using the TenPy library (version 1.1.0) on a local device, in pure Python mode, without any cloud or remote servers. All codes, raw data, and parameter configurations are open-sourced in the GitHub repository (`https://github.com/luoxuejian000/edrn-dmrg-verification`).

- **Round 1 (July 15, 2026):** $L = 40$ single-bond defect spin gap and C calculations (5 data points, script: `game1_defect_scan.py`, output: `data/game1_log.csv`).

- **Round 2 (July 16, 2026):** $L = 60$ single-bond defect spin gap and C calculations (5 data points, scripts: `game1_L60_scan.py` + `game1_L60_fill.py`, output: `data/game1_L60_log.csv`).
- **Round 3 (July 16, 2026):** $L = 40$ and $L = 60$ charge gap calculations (10 data points, script: `game1_charge_gap.py`, output: `data/game1_charge_gap.csv`).

3 Raw Data

No selective reporting. All data points are presented in full.

3.1 Baseline Data: Uniform Open Chain (Defect Strength $1.0t$)

Table 1: Uniform open chain baseline

L	Δ_s	$A = \Delta_s \times L$	C	Charge gap ΔE
40	0.076668	3.0667	0.958885	1.3140
60	0.052320	3.1392	0.964119	1.3318

3.2 $L = 40$ Single-Bond Defect Complete Data

Table 2: $L = 40$ single-bond defect data

Defect strength	Δ_s	$A = \Delta_s \times L$	C	Charge gap ΔE
0.5t	0.136747	5.4699	0.475369	1.2611
0.8t	0.109164	4.3666	0.478021	1.2892
1.0t	0.076668	3.0667	0.479442	1.3140
1.2t	0.047754	1.9102	0.480129	1.3301
1.5t	0.024137	0.9655	0.480227	1.3347

3.3 $L = 60$ Single-Bond Defect Complete Data

Table 3: $L = 60$ single-bond defect data

Defect strength	Δ_s	$A = \Delta_s \times L$	C	Charge gap ΔE
0.5t	0.095642	5.7385	0.479309	1.2987
0.8t	0.077372	4.6423	0.481096	1.3140
1.0t	0.052320	3.1392	0.482060	1.3318
1.2t	0.030031	1.8019	0.482524	1.3440
1.5t	0.013796	0.8277	0.482635	1.3473

4 Data Analysis

This analysis strictly follows the "only diagnose, not prescribe" principle. All statements below are descriptive and do not make normative judgments.

4.1 Monotonicity of A with Defect Strength

$L = 40$: A from 5.47 (0.5t) $\rightarrow$ 4.37 (0.8t) $\rightarrow$ 3.07 (1.0t) $\rightarrow$ 1.91 (1.2t) $\rightarrow$ 0.97 (1.5t). Five data points, all monotonically decreasing, none deviating from the trend.

$L = 60$: A from 5.74 (0.5t) $\rightarrow$ 4.64 (0.8t) $\rightarrow$ 3.14 (1.0t) $\rightarrow$ 1.80 (1.2t) $\rightarrow$ 0.83 (1.5t). Five data points, all monotonically decreasing, none deviating from the trend.

Descriptive statement: A monotonically decreases with increasing defect strength. The weaker the defect (smaller hopping, weaker network connectivity), the larger A ("harder" spin excitation). This monotonic trend is consistently observed across ten data points at two independent L values.

4.2 L-Convergence Test

Table 4: L-convergence comparison

Defect strength	A at $L = 40$	A at $L = 60$	Difference
0.5t	5.4699	5.7385	+4.9%
0.8t	4.3666	4.6423	+6.3%
1.0t	3.0667	3.1392	+2.4%
1.2t	1.9102	1.8019	-5.7%
1.5t	0.9655	0.8277	-14.3%

Descriptive statements:

1. The differences in A between the two L values are within 15%.
2. The monotonic trend is fully consistent across both scales—A decreases monotonically with increasing defect strength.
3. The A(defect) curves retain a stable shape across both scales, without significant L-dependence.

Counterfactual test: If the variation of A were a finite-size transient, the A values at $L = 60$ should regress toward the uniform-chain value (about 3.07–3.14). The data do not show this—$A(0.5t)$ at $L = 60$ is instead larger than at $L = 40$ (5.74 vs. 5.47), and $A(1.5t)$ at $L = 60$ is smaller than at $L = 40$ (0.83 vs. 0.97). The response persists at both scales, and even strengthens in certain regions. This is inconsistent with the finite-size transient hypothesis.

4.3 Constancy of C

Table 5: Variation of C

L	C range	Variation
40	0.4754 – 0.4802	<1%
60	0.4793 – 0.4826	<1%

Descriptive statement: The C value remains almost unchanged at both L values, with a variation below 1%, while A varies by more than 500% (from 0.97 to 5.47). There is no monotonic correspondence between A and C. The variable that A actually tracks is not the global C, but the local bond strength itself, or some local structural change induced by the defect that is not captured by the current definition of C.

4.4 Charge Gap: Gate Condition Test

All 10 data points exhibit clearly positive charge gaps.

Table 6: Charge gap summary

L	Lowest charge gap	Highest charge gap
40	1.2611 (0.5t)	1.3347 (1.5t)
60	1.2987 (0.5t)	1.3473 (1.5t)

Descriptive statement: The charge gap remains nonzero for all defect strengths. The lowest value occurs at 0.5t (the weakest defect) but still exhibits a clear gap of 1.26–1.30t, far above zero. The system is in the Mott insulating phase under all parameters. The dramatic variation of A (from 5.47 to 0.97) occurs inside the Mott insulator, not as a trivial consequence of a metal-insulator transition.

4.5 Functional Form of A vs. Defect Strength

Descriptive statement: A decreases monotonically with increasing defect strength, exhibiting a nonlinear trend. In the weak-defect region (0.5t to 1.0t), the decline of A is relatively gentle. In the strong-defect region (1.0t to 1.5t), the decline is steeper. Near 1.0t (uniform chain), A approaches the uniform-chain baseline value. This nonlinear shape remains consistent across both L values.

5 Conclusions

5.1 Experimental Findings

The monotonic variation of A with local bond strength has been consistently observed across ten data points at two independent L values. In the Mott insulator, the spin-gap prefactor A is not a bulk universal constant, but undergoes a significant, monotonic, nonlinear variation with the local bond strength.

The standard theory prediction (A is a bulk universal constant) does not hold under the parameter range of this experiment. Standard theory (Heisenberg model, Luttinger liquid theory) predicts A should remain unchanged under the uniform-chain assumption, but the data show significant variation of A at all five defect strengths deviating from uniformity, with a variation exceeding 500%.

The phenomenon has passed the L-convergence test. The monotonic trend of A(defect) retains a consistent shape at two independent L values without disappearing as L increases, which does not support the hypothesis that the phenomenon is a finite-size transient.

All data points have passed the gate condition. The charge gap remains nonzero for all defect strengths, and the system stays in the Mott insulating phase. The variation of A occurs inside the Mott insulator and is not a trivial consequence of a metal-insulator transition.

5.2 Theoretical Revision

The original conjecture "A is a function of C" has been revised in this experiment. The experimental data do not support a monotonic relationship between A and the global topological index C—the variation range of C under a single-bond defect is too small (<1%) to allow an effective test of the $A(C)$ relation. What A actually tracks is not the global C, but the local bond strength itself, or some local structural change induced by the defect.

Revised proposition: In the Mott insulator, the spin-gap prefactor A undergoes a significant monotonic variation with the local bond strength. This phenomenon lies beyond the descriptive scope of standard theory, but can be naturally explained from the "relation precedes entity" postulate of the EDRN framework: altering the local structure of the relation network changes the global physical response.

This revision is not a failure of the conjecture, but its refinement through experiment. The physical core of the original conjecture—A is not a bulk universal constant, and changes in the network structure lead to changes in physical response—is supported by the data. What needs revision is the specific mechanistic hypothesis that A tracks topological changes through the variable C.

5.3 Relationship of This Experiment to "Relation Precedes Entity"

This experiment provides the first direct numerical evidence for the philosophical proposition "relation precedes entity" in strongly correlated physics. Standard physics (entity priority) predicts A is a bulk constant; EDRN (relation priority) predicts A varies with local bond strength. The data support the latter.

5.4 Current Limitations

- **Absence of bond-dimension extrapolation:** All data points were computed with $\chi_{\max} = 100$, without multi-χ extrapolation. Independent verification by Drahos will be crucial.
- **Limited L range:** Tests were performed only at $L = 40$ and $L = 60$.
- **Limitations of the C definition:** The current C is insensitive to a single-bond defect (variation <1%).
- **Limited system parameters:** Only $U/t = 4$ and half-filling were tested.
- **Singularity of methodology:** Only TenPy DMRG was used.

5.5 Next Steps

1. Send complete dataset to Drahos for independent verification.

2. Send data to Sultanov for TAT blind testing.

3. Re-examine the definition of C.

4. Extend to periodic chain, larger L, and weak coupling ($U = 0.5$).

6 Data and Script Files

Data files:

- `data/game1_log.csv` – $L = 40$ defect data (5 points)
- `data/game1_L60_log.csv` – $L = 60$ defect data (5 points)
- `data/game1_charge_gap.csv` – Charge gaps (10 points)
- `data/predict1_topology_spin.csv` – Uniform chain baseline

Scripts:

- `game1_defect_scan.py`
- `game1_L60_scan.py`
- `game1_L60_fill.py`
- `game1_charge_gap.py`

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Repository: https://github.com/luoxuejian000/edrn-dmrg-verification